



THE GOOD ACRE

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Unlocking Efficiency Through Strategic Warehouse Location Selection

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Defining the Path to Strategic Warehouse Placement

Strengthening TGA's Mission

The Good Acre (TGA) bridges local farmers and food producers with markets, grocers, and institutions, ensuring their labor's economic value supports their growth and sustains the farming community.



Current Challenge

TGA is exploring 5 potential warehouse locations to improve operations and better serve its mission. However, determining the "best" location is complex due to the multi-dimensional nature of the factors that must be prioritized



Key Questions

- Which factors should be prioritized to align with TGA's mission of supporting farmers and strengthening the supply chain?
- How can we systematically evaluate and rank the locations to make an informed, data-driven decision?



Our Data-Driven Approach to Optimal Location Selection

To determine the best warehouse location, we utilized data from various sources and analyzed it through a structured process.



Data Exploration

- Government census data across counties
- TGA's product and sales records across counties



Factor Identification

- Identified 6 factors that collectively influence the optimal location selection
- Each factor reflects TGA's mission and operational priorities



Weighted Metrics

- Assigned weights to 6 factors based on their importance
- Calculated weighted metric in each location
- Ranked to identify the best fit



Assigning Rank

Locations were ranked based on these final scores, identifying the optimal warehouse location



The 6 Factors Guiding Our Decision

Mean Supplier Distances

Minimizes transportation costs and logistics complexity to the nearest county. **Olmsted** has the shortest average distance to its suppliers and meat producers across various radii.

Top Commodities Sold

Ensures alignment with market demand. **Kandiyohi** leads in sales, contributing nearly **\$1.15 Bn** from top commodities like corn, soybeans, field crops, grain, and turkey.

Gender Proportion

Aligns with TGA's value of inclusivity. **Otter Tail** leads in highest female proportion with **42%**

Total Farm Count

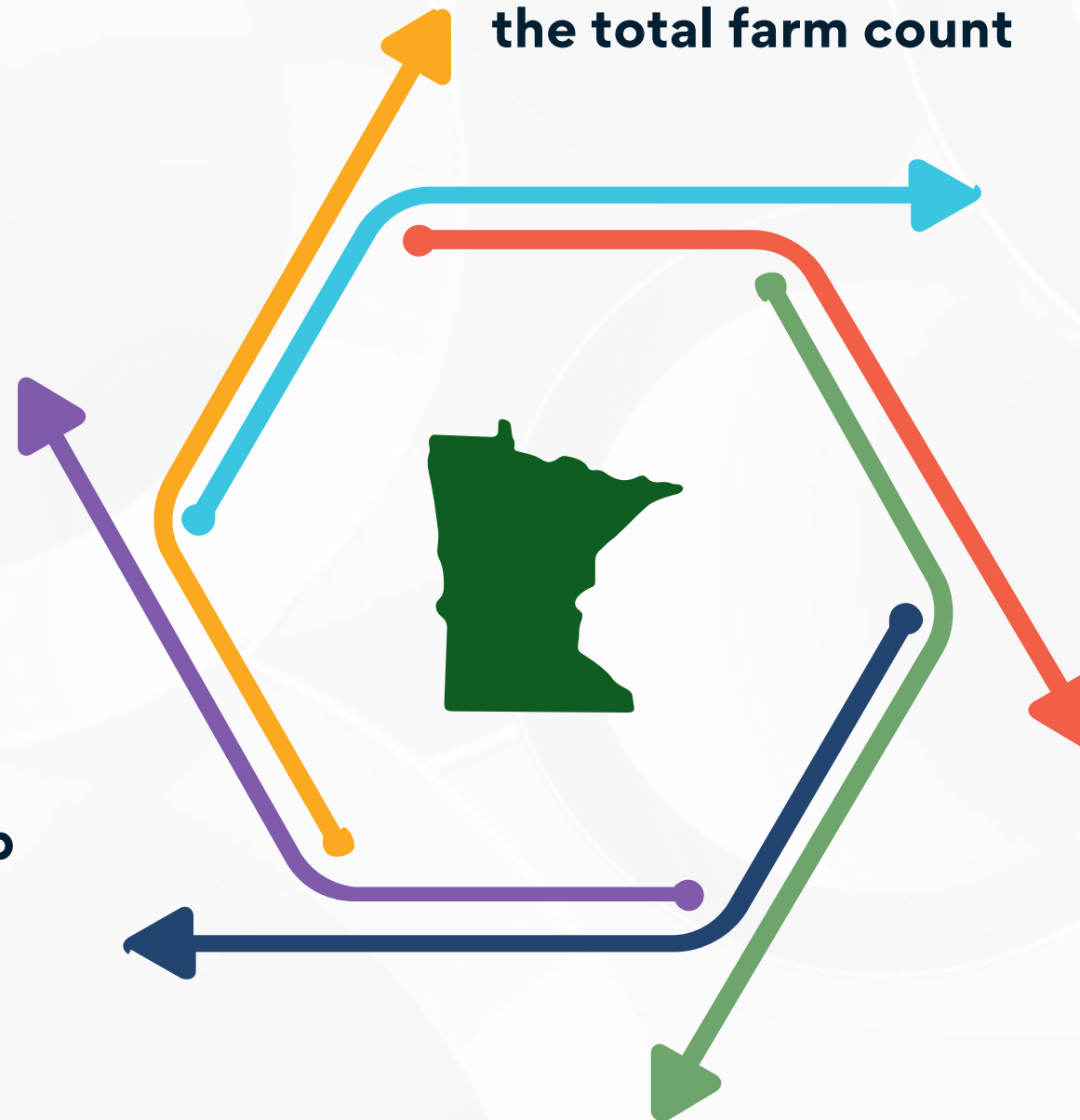
Reflects local farming activity density. **Otter Tail** contributes to about **40%** of the total farm count

Eggs + Vegetable Farm Counts

Supports TGA's most sold items. **Olmsted** and **St. Louis** counties alone account for about **60%** of the total egg and vegetable farms.

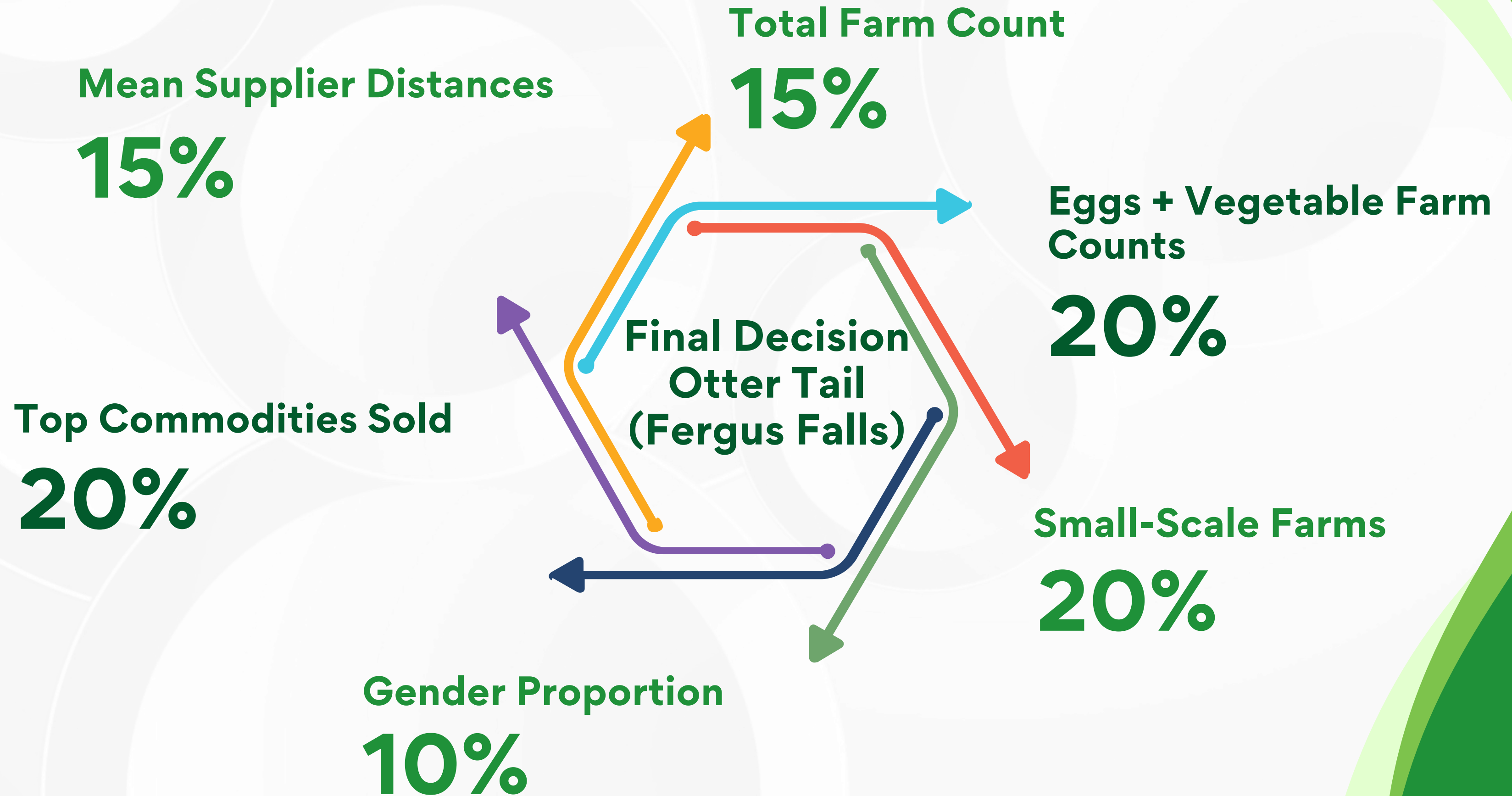
Small-Scale Farms

Prioritizes TGA's mission of economic opportunity. **Otter Tail** ranks highest in the number of small-scale farms, both in terms of area and sales.





Final Selection Based on Weighted Metrics





Recommendations

		Score (5 the highest, 1 the lowest)				
	Weightage	Fergus Falls Otter Tail	Wilmar Kandiyohi	Rochester Olmsted	Duluth St. Louis	Owatonna Steele
Mean Supplier Distance	15%	46.28	33.50	33.25	34.46	40.77
Total Farm Count	15%	3	2	4	5	2
Top Commodity Sold	20%	5	4	3	2	1
Gender Proportion	10%	4	5	2	1	3
Small-Scale Farms	20%	5	4	3	2	1
Eggs + Vegetables Farm Count	20%					
Total		100	19	15	12	8
Rank		1	2	3	4	5



Otter Tail(Fergus Falls): The Ideal Location for TGA's Next Warehouse

With Otter Tail consistently leading in critical factors and never falling below average across all metrics, it is the clear choice for the optimal warehouse location

Small-Scale Farmers:

Ranks #1 in both small-scale farms by area and sales



Female Proportion:

Leads with the highest female farmer proportion at **42%**.



Top Commodities Sales:

Second-highest in commodities sales, contributing **\$1.05B**



Total Farm Count:

Contributes to ~**40%** of the total farm count